

Notes: Mian (JF, 2006)

Distance Constraints: The Limits of Foreign Lending in Poor Economies

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1 Big picture

- **Question.** Can foreign banks solve financial-development problems in poor economies, or do they face distance-related limits when lending to informationally difficult firms?
- **Setting.** The paper studies private corporate lending in Pakistan from **1996 to 2002**, using quarterly loan-level data from the State Bank of Pakistan's credit information system.
- **Core empirical contrast.** Foreign banks and private domestic banks operate in the same country and under the same local regulations, but they differ in the distance between ultimate decision makers and local borrowers or local loan officers.
- **Main conceptual contribution.** The paper distinguishes two explanations for foreign banks' behavior: **distance constraints** versus **risk-preference or prudence differences**. Distance constraints imply that foreign banks are less able to use soft information and relational contracting. Risk-preference stories imply that domestic banks lend to worse borrowers or take excessive risk.
- **Main data advantage.** The data contain roughly **79,323 private-bank loans, 62,253 borrowers, 90 private banks, and 588,546 loan-quarter observations**. The paper observes loan size, borrower characteristics, default, recovery, litigation, bank identity, and ownership links.
- **Main measurement idea.** The paper cannot directly observe soft information. Instead, it uses several proxies for how hard or soft a borrower is: borrower size, borrower city size, whether the borrower is foreign, business-group size, number of creditors, and loan type.
- **Main baseline result.** Foreign banks lend much more to hard-information borrowers. They concentrate lending in very large firms, large cities, foreign firms, conglomerates, firms with multiple creditors, and short-term or transactional loans.
- **Relational-lending result.** Conditional on default, domestic banks are more successful at recovering bad loans and are more likely to resolve distress through bilateral renegotiation rather than litigation.
- **Borrower-selection result.** The recovery and litigation differences survive borrower fixed effects. Even when comparing the same borrower borrowing from both foreign and domestic banks, foreign-bank loans show lower recovery and less relational resolution.
- **Key distance result.** The constraints are stronger for **non-Asian foreign banks** than for **Asian foreign banks**. This supports the interpretation that cultural and geographic distance matter.
- **Main alternative explanations rejected.** The patterns are not explained by bank size, institutional distance, political-enforcement advantages, poor domestic-bank loan quality, lower productivity of domestic-bank borrowers, or related lending.
- **Economic-importance result.** New domestic bank branches after liberalization mostly lend to first-time borrowers and softer firms, suggesting that soft-information firms were not simply waiting to switch from foreign banks. Many appear to have been previously excluded from private credit.

- **Economic takeaway.** Foreign banks may improve stability and provide capital to transparent borrowers, but they are not a full substitute for domestic financial institutions. In poor economies, domestic banks can be essential for lending to small, local, opaque, relationship-dependent firms.

2 Data and measurement (institutional context and key objects)

2.1 Why Pakistan is a useful laboratory

- Pakistan provides a setting where foreign and domestic private banks operate in the same local market.
- The banking sector was nationalized in the early 1970s, but foreign banks continued to operate. After reforms beginning in 1991, new private domestic banks and foreign banks were allowed to enter and expand.
- By the start of the sample period, private domestic and foreign banks had similar aggregate private-bank market shares.
- Both foreign and domestic private banks were subject to the same prudential regulations and banking rules.
- This helps isolate differences related to bank organization and distance rather than obvious legal restrictions on foreign banks.

Interpretation. The setting is attractive because the paper compares foreign and domestic private banks inside the same institutional environment. The main difference is not the borrower country, but the bank's ownership, organizational distance, and familiarity with local soft information.

2.2 Sample construction

- The raw data cover quarterly loan-level information on corporate loans outstanding in Pakistan over **25 quarters**, from April 1996 to April 2002.
- A loan is defined as a unique **bank–borrower pair**. If a facility is renewed over time, it is still treated as the same loan relationship.
- The paper excludes government-bank lending and loans to government firms, focusing on private-sector business loans made by private banks.
- The final cross-sectionalized sample contains **79,323 loans**, **62,253 unique borrowers**, and **90 private banks**.
- The panel version contains **588,546 loan-quarter observations**.
- Variables are converted into real 1995 Pakistani rupees and then averaged over time when the paper uses the cross-sectional loan as the unit of observation.

Interpretation. Most of the analysis is at the loan-relationship level. The time dimension is used where needed, but the central question is whether the same credit market is segmented by bank type.

2.3 Foreign and domestic bank classification

- Banks are classified according to the location of the controlling shareholders.
- The sample contains **22 foreign banks** and **68 private domestic banks**.
- Foreign banks are headquartered in Europe, the Middle East, the United States, and Japan.
- In the sample, foreign banks account for approximately **49%** of private-bank lending and domestic banks for approximately **51%**.

Interpretation. Foreign banks are not marginal players. They are large enough that differences in their lending behavior matter for the allocation of credit in the economy.

2.4 Meaning of distance

The paper distinguishes three possible distance concepts:

1. **Geographic or cultural distance.** Distance between the foreign bank's headquarters and local operations in Pakistan. This is correlated with language, culture, social networks, and familiarity with local conditions.
2. **Hierarchical distance.** Distance inside the bank due to layers of organization. Larger banks may be more hierarchical and may rely more on hard information.
3. **Institutional distance.** Difference between the legal, regulatory, and institutional environment of the bank's home country and Pakistan.

Interpretation. The paper's central distance concept is not the distance between the borrower and the branch officer. It is the distance between the bank's controlling decision makers and the local loan officers or local lending environment.

2.5 Hard versus soft information

- **Hard information** is verifiable, codifiable, and easier to transmit across distance. Examples include firm size, credit history, formal foreign ownership, and observable business-group affiliation.
- **Soft information** is difficult to verify and communicate. Examples include managerial ability, trustworthiness, local reputation, business potential, and informal knowledge acquired through repeated interactions.
- Relationship lending depends heavily on soft information because the lender must monitor the borrower, renegotiate when necessary, and recover value in weak legal environments.

Interpretation. Foreign banks can lend effectively when the necessary information is standardized and verifiable. They have more difficulty when the loan depends on trust, local knowledge, and discretionary judgment by the local loan officer.

2.6 Six proxies for hard-information borrowers

1. **Borrower size.** Larger firms are more visible and more likely to have formal records, reputation, and collateral.
2. **Borrower location.** Firms in large cities are easier to observe and monitor, and their information is more standardized.
3. **Foreign firm dummy.** Multinational or foreign-owned firms are easier for foreign banks to evaluate and may already be part of international networks.
4. **Business-group size.** Larger groups and conglomerates have reputations, internal capital markets, and more public visibility.
5. **Number of creditors.** A borrower with several creditors has a credit history and can be partially evaluated through the behavior of other banks.
6. **Loan type.** Longer-term fixed-capital loans are more relationship-intensive; short-term working-capital loans, letters of credit, and guarantees are more transactional.

Interpretation. Each proxy is imperfect, but together they create a consistent picture of whether foreign banks concentrate on borrowers that require less soft information.

2.7 Main outcome variables

- **Lending composition.** The paper compares the types of borrowers funded by foreign and domestic banks.
- **Default rate.** Default is defined as nonpayment for more than 90 days.
- **Recovery.** Conditional on default, the paper asks whether the bank recovers any amount from the defaulted loan and what the recovery rate is.
- **Litigation.** Conditional on default, the paper asks whether the bank takes the borrower to court rather than resolving distress through bilateral renegotiation.
- **Loan return.** The paper combines interest rates, default rates, and recovery rates to compare whether domestic and foreign loans have similar expected returns.
- **Real borrower quality.** For textile firms, the paper uses export outcomes as a nonfinancial measure of borrower productivity.

Interpretation. The paper does not rely only on who foreign banks lend to. It also studies what happens after a loan deteriorates, because renegotiation and recovery are central relational-lending tasks.

2.8 Special variables for mechanism tests

- **Asian versus non-Asian foreign banks.** Foreign banks are split into 11 Asian banks and 11 non-Asian banks. Non-Asian banks are more geographically and culturally distant from Pakistan.
- **Political connectedness.** A borrower is politically connected if one of its directors ran in the 1993 or 1997 national elections. About **19%** of loans are classified as politically connected.
- **Bank size.** The paper uses both local size in Pakistan and global size from worldwide assets.
- **Institutional-distance measures.** The paper considers legal origin, regulatory-distance measures, and income distance between Pakistan and a foreign bank's home country.
- **New branch lending.** The paper studies loans made by bank branches opened during the sample period to evaluate whether new domestic banks expand credit to borrowers previously excluded from private credit.

Interpretation. These variables allow the paper to move from a descriptive foreign-versus-domestic comparison to tests of why foreign banks behave differently.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Conceptual framework: distance constraints versus risk preferences

- **Distance-constraints hypothesis.** Foreign banks have higher costs of acquiring, verifying, communicating, and acting on soft information. Therefore, they avoid soft-information firms and perform worse at relationship-intensive tasks such as renegotiation and recovery.
- **Risk-preference hypothesis.** Domestic banks lend to soft-information borrowers because they are less prudent, take excessive risk, or make low-quality loans. Foreign banks avoid these borrowers because foreign banks have higher standards.
- The paper's empirical design looks for outcomes where these theories make different predictions.

Interpretation. The key issue is not merely whether foreign banks and domestic banks lend to different borrowers. The key issue is whether domestic banks are better at relational lending or simply less disciplined.

3.2 Baseline foreign-bank comparison

Let Y_{ij} denote an outcome for borrower i borrowing from bank j , such as default, recovery, litigation, or a hard-information proxy. Let F_j equal one if bank j is foreign.

$$Y_{ij} = \alpha + \beta F_j + \varepsilon_{ij}. \quad (1)$$

Interpretation. The coefficient β measures the average difference between foreign and domestic banks. If Y is a hard-information proxy, a positive β means foreign banks lend to harder-information borrowers. If Y is recovery, a negative β means foreign banks recover less conditional on default.

3.3 Borrower controls

The main controlled specification adds borrower and loan characteristics:

$$Y_{ij} = \alpha + \beta F_j + X_i' \gamma + \varepsilon_{ij}, \quad (2)$$

where X_i includes city dummies, industry dummies, loan-type dummies, borrower-size dummies, group-size dummies, number-of-creditors dummies, and a foreign-firm indicator.

Interpretation. This specification asks whether foreign–domestic differences remain after comparing borrowers that are similar on observed characteristics. The controls are deliberately rich and nonparametric, so they likely absorb many relevant borrower differences.

3.4 Borrower fixed effects

For borrowers that borrow from both foreign and domestic banks, the paper estimates:

$$Y_{ij} = \alpha_i + \beta F_j + \varepsilon_{ij}, \quad (3)$$

where α_i is a borrower fixed effect.

Interpretation. This is the cleanest borrower-selection test. The estimate compares foreign and domestic loans to the same borrower. If foreign banks recover less from the same distressed borrower, the result cannot be explained by foreign banks lending to different types of borrowers.

3.5 Standard errors and weighting

- Standard errors are clustered at the **bank level**, because loan outcomes are correlated within banks.
- The paper reports both unweighted and loan-size-weighted results because the distribution of loan size is highly skewed.
- Weighting asks whether the results matter economically in rupee terms, not only in the count of loan relationships.

Interpretation. The empirical design is careful about two common issues: correlated errors within banks and the possibility that many small loans dominate loan-count regressions.

3.6 Distance-interaction specification

Let Z_j be a distance attribute of a foreign bank, such as being non-Asian, having large global size, or being institutionally distant from Pakistan. The interaction specification is:

$$Y_{ij} = \alpha + \beta_1 F_j + \beta_2 (F_j \times Z_j) + X_i' \gamma + \varepsilon_{ij}. \quad (4)$$

Interpretation. If a specific form of distance is the source of the constraint, the foreign-bank difference should be larger for foreign banks with more of that distance. In the paper, this logic is most important for the Asian versus non-Asian split.

3.7 Political-enforcement difference-in-differences

To test whether domestic banks are better because they can use political pressure to enforce loans, the paper estimates:

$$Y_{ij} = \alpha + \beta_1 DOM_j + \beta_2 UNCON_i + \beta_3 (DOM_j \times UNCON_i) + \varepsilon_{ij}, \quad (5)$$

where DOM_j equals one for domestic banks and $UNCON_i$ equals one for politically unconnected borrowers.

Interpretation. If domestic banks have a political-enforcement advantage, they should do especially well against politically unconnected firms. For default, this would imply a negative interaction; for recovery, it would imply a positive interaction. The paper does not find a pattern supporting this channel.

3.8 Loan-return calculation

The paper evaluates whether domestic-bank loans are poor-quality loans by comparing loan returns. If d_j is the default rate, r_j is the loan interest rate, and q_j is the recovery fraction conditional on default, then:

$$1 + R_j = (1 - d_j)(1 + r_j) + d_j q_j. \quad (6)$$

Interpretation. Domestic banks charge higher interest rates and have somewhat higher raw default rates, but they also recover more in default. The paper finds that the implied loan return is about the same for domestic and foreign banks.

3.9 New-branch entry test

- The paper studies loans made by branches opened after 1996.

- It asks whether new domestic branches mainly take borrowers away from existing foreign banks or instead lend to first-time borrowers.
- If distance constraints are small, many soft-information firms should have been borrowing from foreign banks and then switched to domestic banks when domestic branches entered.
- If distance constraints are large, new domestic branches should mostly create access for borrowers that foreign banks had not been serving.

Interpretation. The new-branch test gives the paper a way to estimate the economic importance of distance constraints, not just their statistical significance.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the setting helps identification

- Foreign and domestic banks operate in the same country, same borrower market, and same local regulatory environment.
- Foreign banks had been present in Pakistan before the rise of private domestic banks, so the results cannot be explained simply by foreign banks being new entrants unfamiliar with Pakistan.
- Foreign and domestic private banks have roughly equal aggregate market shares, so the comparison is between economically important bank groups.

Economic message. The institutional setting reduces many obvious confounds. The paper does not compare foreign banks in one country with domestic banks in another country; it compares them in one market.

4.2 Borrower-selection concerns

- A major concern is that foreign banks may lend to different borrowers, and those borrowers may be intrinsically different in unobserved ways.
- The paper first adds a large set of borrower controls.
- It then uses borrower fixed effects for borrowers that borrow from both foreign and domestic banks.
- The borrower-fixed-effect results are especially important for recovery and litigation because they compare outcomes for the same borrower.

Economic message. The paper's most credible evidence is not only that foreign banks lend to different borrowers. It is that foreign banks perform differently at relational tasks even for borrowers who also borrow from domestic banks.

4.3 Bank-level selection concerns

- Another concern is that foreign banks may differ from domestic banks in size, age, risk appetite, home-country regulation, or organizational structure.
- The paper tests global and local bank size. Size does not explain the main geographic/cultural-distance results.
- The paper also tests institutional-distance measures: legal origin, regulatory distance, and income distance. These do not explain the patterns.
- Bank age is unlikely to explain the results because foreign banks had been present longer, and the patterns remain when comparing domestic banks with newer foreign banks.

Economic message. The core result is not simply a large-bank versus small-bank result or a developed-country-regulation result. The evidence points more directly to cultural and geographic distance.

4.4 Political-enforcement concerns

- In weak legal environments, domestic banks might have informal enforcement advantages through political or social networks.
- The paper tests whether domestic banks do especially well against politically unconnected borrowers.
- The evidence does not support this enforcement interpretation.

Economic message. The domestic-bank advantage appears more informational and relational than coercive or political.

4.5 Risk-preference concerns

- The risk-preference view says domestic banks lend to soft-information firms because they are reckless, make bad loans, or engage in related lending.
- The paper rejects this by showing similar loan returns for domestic and foreign banks.
- It also shows that domestic-bank borrowers in textiles are not less productive in export data.
- Related lending by domestic private banks is low and does not appear to drive the findings.

Economic message. Domestic banks are not simply making bad loans that foreign banks wisely avoid. They appear able to serve borrowers that foreign banks have difficulty serving.

4.6 What the paper can and cannot prove

- The paper cannot randomly assign banks to be foreign or domestic.

- It also cannot directly observe soft information, culture, trust, or internal bank communication costs.
- However, the consistency across lending composition, recovery, litigation, borrower fixed effects, Asian versus non-Asian banks, and new-branch entry makes the distance-constraint interpretation persuasive.

Economic message. The empirical design is not a randomized experiment, but it is a strong accumulation of evidence. The same mechanism explains many different patterns.

5 Tables and figures

5.1 Figure 1: Defining distance

Read:

- The figure distinguishes the distance between a foreign bank's headquarters and the local loan officer from the distance between the loan officer and the borrower.
- The paper focuses on the first distance: the principal or CEO is abroad, while the loan officer and borrower are in Pakistan.
- Distance can be geographic, cultural, institutional, or hierarchical.

Economic message. The paper is about the internal organizational problem of foreign lending. The challenge is not only that borrowers are far from branches; it is that local soft information must travel through a distant organization.

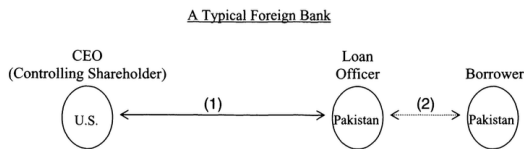


Figure 1. Defining distance. Here distance can refer to geographical, cultural, institutional, or hierarchical distance.

Table I
Data Description

Our sample includes all private sector business loans given out by private banks between 1996 and 2002. Panel A reports the basic dimensions of our quarterly panel data. A "loan" is defined as a bank-borrower pair. In other words if a loan facility for a firm is renewed over time, it still counts as a single loan. The number of borrowers in Panel A is smaller than the number of loans as a single borrower may borrow from multiple banks. "Loans" in Panel B refer to the average of total outstanding loans from 1996 to 2002 in billions of 1996 Pak Rs.

Panel A: Data Coverage			
No. of quarters	25 (April 1996–April 2002)		
No. of banks	90		
No. of unique borrowers	62,253		
No. of unique loans	79,323		
No. of loan-quarter observation	588,546		
	Mean	Min	Max
No. of loans in a quarter	24,716	15,952	31,727
No. of banks in a quarter	85.2	78	88
Panel B: Bank Classification			
	No. of Banks	Loans	Market Share
Foreign	22	260	49%
Private domestic	68	268	51%

5.2 Table I: Data description

Read:

- The sample covers **25 quarters** from April 1996 to April 2002.

Table II
Summary Statistics

The table presents summary statistics for the variables used in this paper. Panel A gives the distribution of the six "hard information" proxies used in this paper. The distribution is given both in terms of the percentage of lending falling into a given category, as well as the number of loans. "Size" is defined as the total borrowing of a firm from the financial sector. "Location" refers to the size of the city/town the borrower is located in: *small* refers to a town with less than 0.5 million people, *medium* with population between 0.5 and 2 million, and *large* with a population of over 2 million. "Foreign" captures whether a borrowing firm is foreign-owned or not. "Group Size" refers to the number of firms in the group that a firm belongs to: *stand alone* firms do not belong to any group, *intermediate* firms belong to groups of size 2–50, and *conglomerate* firms belong to groups of more than 50. Panel B reports the cross-correlations between the six variables in Panel A. In computing the cross-correlations, "Size" is coded as 1–5 for the five size categories, "Location" is coded as 1–3 for the three city/town sizes, "Foreign Firm" is a dummy variable, "Group Size" is coded 1–3 for the three group sizes, and "Loan Type" is coded as 0/1 with "1" representing if the loan is a short-term loan (i.e., not a fixed loan). Panel C reports the summary statistics of loan level financial variables. Note that the litigation and recovery information is only available from April 2001 to April 2002.

Panel A: "Hard Information" Variables					
Size (Percentile)	0–50	50–75	75–95	95–99	99–100
Percentage of total lending	1%	3%	11%	20%	65%
No. of loans (79,323 total)	23,854	19,632	21,004	8,484	6,349
Location	Small	Medium	Large	Unclassified	
Percentage of total lending	2%	8%	89%	1%	
No. of loans (79,323 total)	5,876	10,918	58,952	3,577	
Foreign Firm	No				Yes
Percentage of total lending	82%				18%
No. of loans (79,323 total)	78,830				493
Group Size	Stand-alone	Intermediate	Conglomerate	Unclassified	
Percentage of total lending	12%	18%	66%	3%	
No. of loans (79,323 total)	35,510	15,504	11,801	16,508	
Number of creditors	1	2	3	4	5
Percentage of total lending	13%	7%	6%	5%	3%
No. of loans (79,323 total)	47,319	12,152	5,388	3,022	1,783
				1,783	9,659
Loan Type	Fixed	Working Capital	Letter of Credit	Guarantees	Other
Percentage of total lending	10%	59%	13%	8%	10%
No. of loans (79,323 total)	20,806	42,632	9,000	5,584	1,301

(continued)

- It contains **90 private banks**, **62,253 borrowers**, **79,323 loans**, and **588,546 loan-quarter observations**.
- Foreign banks account for **22 banks** and about **49%** of private-bank lending.
- Domestic private banks account for **68 banks** and about **51%** of private-bank lending.

Economic message. Foreign banks are large enough that their lending constraints have aggregate importance. The paper is not studying a small peripheral sector.

5.3 Table II: Hard-information proxies and summary statistics

Read:

- The paper defines borrower size categories, city-size categories, foreign-firm status, group-size categories, number of creditors, and loan type.
- Large firms dominate total lending: the top 1% by size account for a large share of total lending.
- The average loan size is about **6.7 million** 1995 Pakistani rupees.
- The overall default rate is around **4.5%** in the unweighted data and **4.8%** in the weighted data.
- Conditional on default, banks litigate roughly one-third of the time and recover at least some amount in roughly 40% of cases.

Economic message. The data allow the paper to study not only lending access but also distressed-loan resolution. This is crucial because relational contracting matters most when something goes wrong.

5.4 Figure 2 and Table III: Lending composition of foreign and domestic banks

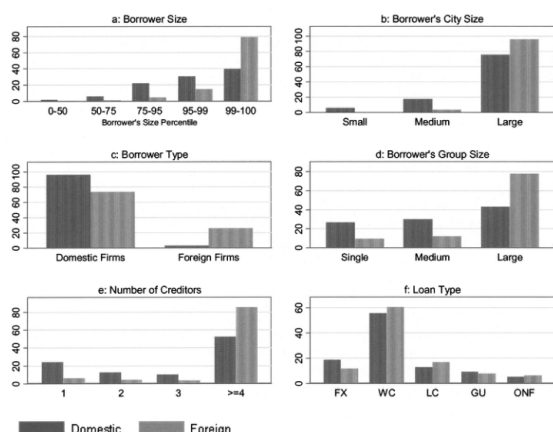


Figure 2. Lending composition of domestic and foreign banks. Each bar in the figure above represents the percentage of overall bank lending given to a particular category. Figure 2a gives lending composition by borrower size percentile, Figure 2b by size of borrower city, Figure 2c by foreign/local firms, Figure 2d by borrower's group size, Figure 2e by borrower's number of creditors, and Figure 2f by borrower's loan type. The loan type categories in Figure 2f are: "FX" for fixed loans, "WC" for working capital, "LC" for letter of credit, "GU" for guarantees, and "ONF" for other nonfunding loans. All loan types are short-term loans except "FX," which have a maturity greater than 2 years.

Table III
Lending Composition: Differences between Domestic and Foreign Banks

The results are based on the cross-sectional database of 79,323 loans covering a period of 1996–2002. Standard errors reported in parentheses are clustered at the bank-level (90 banks in all). Size is coded 1 through 5, representing the five borrower-size classifications shown in Table II; Location is coded 1 through 3, representing the city size of the borrower as in Table II; Foreign Firm is a dummy for whether the borrower is a multinational; Group Size is coded 1 through 3, representing the group size of the borrower, and Loan Type is a dummy for whether the loan is a short-term loan.

	Value-Weighted Means for "Hard Information" Proxies					
	(1) Size	(1) Location	(3) Foreign Firm	(4) Group Size	(5) No. of Creditors	(6) Loan Type
Foreign	4.70 (0.06)	2.95 (0.02)	0.26 (0.03)	2.72 (0.04)	3.68 (0.08)	0.93 (0.03)
Domestic	4.01 (0.06)	2.72 (0.05)	0.035 (0.005)	2.23 (0.05)	2.92 (0.08)	0.81 (0.11)
Difference	0.70 (0.07)	0.24 (0.05)	0.23 (0.03)	0.50 (0.06)	0.77 (0.11)	0.12 (0.11)
Difference with controls ^a	0.13 (0.03)	0.18 (0.03)	0.11 (0.03)	0.11 (0.04)	0.08 (0.05)	0.16 (0.12)
Observations	79,323	75,746	79,323	62,815	79,323	79,323

^aThe controls include 5 loan-type dummies, 5 borrower size dummies, a dummy for whether the borrower is a foreign firm, a dummy for the number of creditors the borrower has, 3 group size dummies, 134 dummies for each of the city/town of borrower, and 21 dummies for the industry of the borrower.

Read:

- Foreign banks lend much more to the largest firms: more than 80% of their lending goes to the top 1% by size, compared with less than 40% for domestic banks.
- Foreign-bank lending is almost entirely concentrated in the largest cities.
- Foreign banks lend much more to foreign firms than domestic banks do.
- Foreign banks lend more to large conglomerates and to firms with several creditors.
- Table III confirms these patterns statistically. The value-weighted foreign–domestic difference is positive for borrower size, location, foreign-firm status, group size, number of creditors, and loan type.
- Many differences remain significant even after controlling for the other hard-information proxies.

Economic message. Foreign banks specialize in borrowers whose information is easier to verify and transmit. Domestic banks are the main lenders to softer, more local, more relationship-dependent borrowers.

5.5 Figure 3 and Table IV: Default, recovery, and litigation

Read:

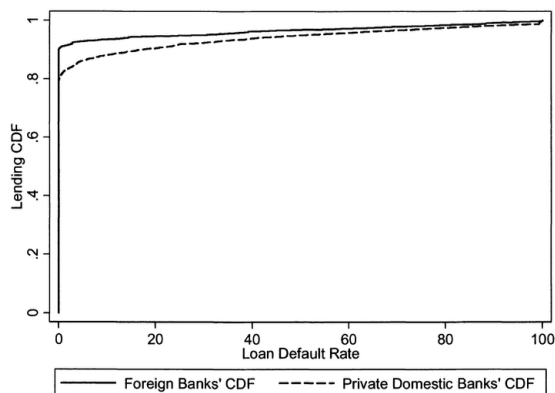


Figure 3. CDF of bank lending by default rate. The figure plots the cumulative distribution function of bank lending against the loan default rate. A point on the graph represents the percentage of aggregate lending with a default rate at or below the corresponding default rate on the x-axis. The upper line represents the CDF graph for foreign banks, while the lower line represents the CDF for domestic banks.

Table IV
Default, Recovery, and Litigation

The results are based on the cross-sectional database of 79,323 loans covering a period of 1996–2002. Standard errors given in parentheses are clustered at the bank-level (90 banks in all). The recovery and litigation data cover the period April 2001 – April 2002.

	Conditional on Default					
	Mean Default Rate (%)		Mean Recovery Rate (%)		Mean Litigation Rate (%)	
	(1)	(2)	(3)	(4)	(5)	(6)
Foreign	3.53 (0.81)	4.32 (0.77)	18.11 (3.86)	18.96 (2.93)	38.87 (8.50)	63.18 (5.08)
Domestic	6.04 (1.98)	4.54 (1.00)	44.39 (7.33)	46.05 (10.62)	33.39 (2.28)	24.2 (5.35)
Difference	-2.52 (2.13)	-0.22 (1.25)	-26.28 (8.23)	-27.09 (10.98)	5.47 (8.63)	38.97 (7.30)
Difference with controls ^a	-2.54 (2.08)	-0.67 (1.23)	-21.0 (6.92)	-18.08 (5.99)	5.81 (7.66)	34.06 (7.70)
Weighted	Yes	No	Yes	No	Yes	No
Observations	79,323	79,323	5,762	5,762	5,762	5,762

^aThe controls include 5 loan-type dummies, 5 borrower size dummies, dummy for whether the borrower is a foreign firm, dummy for the number of creditors the borrower has, 3 group size dummies, 134 dummies for each of the city/town of borrower, and 21 dummies for the industry of the borrower.

- Raw default rates are somewhat lower for foreign banks than for domestic banks, but the difference is not statistically decisive.
- Conditional on default, foreign banks recover much less. In Table IV, foreign banks recover at least some amount on about **18–19%** of defaulted loans, while domestic banks recover on about **44–46%**.
- Foreign banks are more likely to litigate in unweighted results: about **63%** of foreign-bank defaulted loans go to litigation, compared with about **24%** for domestic banks.
- The litigation difference is weaker in loan-size-weighted results, suggesting foreign banks may behave differently for larger defaults.
- The recovery difference is economically large and robust to extensive controls.

Economic message. Foreign banks' problem is not simply avoiding risk. They appear less effective at relationship-intensive distress resolution, especially recovery after default.

5.6 Table V: Borrower fixed effects

Read:

- Table V restricts the sample to borrowers that borrow from both foreign and domestic banks.
- With borrower fixed effects, there is no significant foreign-bank difference in default rates.
- Conditional on default, foreign-bank loans still have lower recovery: roughly **13–14 percentage points** lower in the reported specifications.
- Foreign banks also show more litigation in the unweighted borrower-fixed-effect comparison.

Economic message. The recovery result is not just borrower selection. For the same borrower, domestic banks appear better at relational resolution after default.

Table V
Default, Recovery, and Litigation Using Borrower Fixed Effects
The results are based on the cross-sectional database of 15,453 loans from 1996 to 2002. These are loans taken by firms that borrowed from both foreign and local banks. The recovery and litigation data cover the period April 2001–April 2002. Columns (1) and (2) have 4,217 fixed effects (one for each borrower lending from both domestic and foreign banks). Columns (3)–(6) have 101 fixed effects (one for each borrower lending from both domestic and foreign banks, and defaulting on both). Standard errors given in parentheses are clustered at the bank level (90 banks in all).

	Conditional on Default					
	Default Rate (%)		Recovery Rate (%)		Litigation Rate (%)	
	(1)	(2)	(3)	(4)	(5)	(6)
Foreign	0.47 (0.72)	0.16 (0.59)	−14 (7.30)	−13.24 (4.33)	5.2 (9.40)	13.39 (5.96)
Loan-size weighted	Yes	No	Yes	No	Yes	No
Borrower fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations R^2	15,453 0.71	15,453 0.59	390 0.63	390 0.47	390 0.55	390 0.49

Table VI
Is Distance Geography/Culture?

Panel A reports differences in lending composition between domestic, Asian foreign, and non-Asian foreign banks. The results are based on the cross-sectional database of 79,323 loans covering a period of 1996–2002. Size is coded 1 through 5, representing the five borrower size classifications shown in Table II; Location is coded 1 through 3, representing the city size of the borrower as in Table II; Foreign Firm is a dummy for whether the borrower is a multinational; Group Size is coded 1 through 3, representing the group size of the borrower; and Loan Type is a dummy for whether the loan is a short-term loan. Panel B reports differences between the three types of banks in their recovery and litigation rates once a loan goes into default. The omitted category in Panel B is Asian foreign banks. Regressions in Panel B include 5 loan-type dummies, 5 borrower size dummies, a dummy for whether the borrower is a foreign firm, a dummy for the number of creditors the borrower has, 3 group size dummies, 134 dummies for each of the city/town of borrower, and 21 dummies for the industry of the borrower. The recovery and litigation data cover the period April 2001–April 2002. Both panels report standard errors in parentheses clustered at the bank level (90 banks in all).

Panel A: Lending Composition—Value-Weighted Means for “Hard Information” Proxies						
	Size (1)	Location (2)	Foreign Firm (3)	Group Size (4)	No. of Creditors (5)	Loan- Type (6)
(Domestic − Asian foreign)	−0.52 (0.07)	−0.16 (0.06)	−0.13 (0.04)	−0.36 (0.08)	−0.56 (0.11)	−0.12 (0.12)
Asian foreign	4.53 (0.06)	2.87 (0.04)	0.16 (0.04)	2.58 (0.07)	3.46 (0.08)	0.93 (0.04)
(Non-Asian foreign − Asian foreign)	0.21 (0.08)	0.10 (0.04)	0.12 (0.05)	0.17 (0.08)	0.27 (0.11)	0.01 (0.05)
Observations	79,323	75,746	79,323	62,815	79,323	79,323

Panel B: Recovery and Litigation				
	Recovery		Litigation	
	(1)	(2)	(3)	(4)
Non-Asian foreign	−10.84 (8.56)	−14.99 (6.30)	0.26 (0.07)	0.14 (0.11)
Domestic	11.52 (8.13)	17.13 (7.55)	−0.18 (0.08)	−0.02 (0.08)
Loan-size weighted	No	Yes	No	Yes
Other controls	Yes	Yes	Yes	Yes
Observations	5,762	5,762	5,762	5,762
R^2	0.19	0.22	0.21	0.13

5.7 Table VI: Is distance geography or culture?

Read:

- Foreign banks are split into **Asian** and **non-Asian** banks.
- Asian foreign banks are closer to Pakistan geographically and culturally than banks from the United States or Europe.
- Domestic banks lend to softer borrowers than Asian foreign banks.
- Non-Asian foreign banks lend to even harder-information borrowers than Asian foreign banks.
- Conditional on default, non-Asian foreign banks recover less than Asian foreign banks, while domestic banks recover more.
- Non-Asian foreign banks also litigate more than Asian foreign banks in the reported specifications.

Economic message. The foreign-bank constraint becomes stronger as cultural and geographic distance increases. This is one of the paper’s most important mechanism results.

5.8 Table VII: Political enforcement

Read:

- The paper asks whether domestic banks do better because they can use political or informal pressure.

Table VII
Are Distance Constraints Driven by Political Enforcement?

The results are based on the cross-sectional database of 79,323 loans covering a period of 1996–2002. Standard errors in parentheses are clustered at the bank-level (90 banks in all). A loan is classified as “politically connected” if the firm taking the loan has a director who stood in one of the national elections of 1993 or 1997. The recovery data covers the period April 2001 to April 2002. Regression includes a constant as well.

	(%)		Conditional on Default Recovery Rate (%)	
	(1)	(2)	(3)	(4)
Domestic	1.09 (2.10)	−0.7 (1.89)	24.77 (7.78)	19.47 (8.49)
Politically unconnected	−2.16 (0.45)	−2.8 (0.65)	−1.77 (9.92)	−3.56 (4.13)
Domestic * politically unconnected	2.18 (0.53)	0.92 (0.84)	3.06 (11.81)	10.43 (6.56)
Loan-size weighted	Yes	No	Yes	No
R^2	0.01	0	0.04	0.04
Observations	79,323	79,323	5,762	5,762

- A borrower is politically connected if a director ran in the 1993 or 1997 national elections.
- The test compares connected and unconnected borrowers across domestic and foreign banks.
- The interaction terms do not support the view that domestic banks have a special political-enforcement advantage against unconnected firms.

Economic message. The evidence points away from a coercive political-enforcement story. Domestic banks seem to have a relational-information advantage rather than simply better access to political pressure.

5.9 Table VIII and Figure 4: Is distance really bank size?

Read:

- The paper controls for worldwide bank size and allows flexible effects for banks in the top half of the size distribution.
- The non-Asian foreign-bank patterns remain after controlling for global bank size.
- Local-size controls produce similar conclusions.
- Figure 4 shows the distribution of local and global bank size for foreign and domestic banks.

Economic message. The distance result is not simply a large-bank bureaucracy result. Bank size can matter in principle, but it does not explain the main empirical patterns here.

5.10 Institutional-distance tests

Read:

- The paper also considers legal distance, regulatory distance, and income distance between Pakistan and the foreign bank’s home country.
- These institutional-distance measures do not explain the results, and including them does not remove the geographic/cultural-distance effect.

Table VIII
Is Distance Size?

Panel A reports differences in lending composition between domestic, Asian foreign, and non-Asian foreign banks after controlling for world-wide bank size. This "bank size" is calculated by summing up the total assets of a bank from all of its branches the world over (data source: BankScope). "Big" is a dummy variable that is 1 if the bank belongs to the top half of the "bank size" distribution. Size is coded 1 through 5, representing the five borrower-size classifications shown in Table II; Location is coded 1 through 3, representing the city size of the borrower as in Table II; Foreign Firm is a dummy for whether the borrower is a multinational; Group Size is coded 1 through 3, representing the group size of the borrower; and Loan Type is a dummy for whether the loan is a short-term loan. Panel B reports differences between the three types of banks in their recovery and litigation rates after controlling for bank size as in Panel A above. The omitted category in Panel B is Asian foreign banks. Regressions in Panel B include 5 loan-type dummies, 5 borrower size dummies, a dummy for whether the borrower is a foreign firm, a dummy for the number of creditors the borrower has, 3 group size dummies, 134 dummies for each of the city/town of borrower, and 21 dummies for the industry of the borrower. The recovery and litigation data cover the period April 2001–April 2002. Both panels report standard errors in parentheses clustered at the bank level (90 banks in all).

Panel A: Lending Composition—Value-Weighted Means for "Hard Information" Proxies						
	Size (1)	Location (2)	Foreign Firm (3)	Group Size (4)	No. of Creditors (5)	Loan- Type (6)
Domestic Bank Dummy	-0.59 (0.08)	-0.20 (0.06)	-0.14 (0.05)	-0.41 (0.08)	-0.73 (0.09)	-0.01 (0.07)
Non-Asian Foreign Bank Dummy	0.52 (0.16)	0.25 (0.07)	0.19 (0.09)	0.38 (0.15)	0.93 (0.20)	-0.37 (0.27)
Log of Bank Size	0.01 (0.24)	-0.01 (0.02)	0.00 (0.00)	0.20 (0.11)	-0.10 (0.19)	-0.05 (0.15)
Log of Bank Size * Big	-0.09 (0.25)	-0.02 (0.03)	-0.02 (0.02)	-0.25 (0.11)	-0.06 (0.19)	0.14 (0.16)
Big	1.87 (3.02)	0.25 (0.36)	0.31 (0.34)	3.26 (1.47)	1.40 (2.39)	-1.61 (2.26)
Constant	4.03 (2.97)	3.31 (0.29)	0.14 (0.05)	0.23 (1.37)	5.02 (2.26)	0.86 (1.90)
Observations	79,323	75,746	79,323	62,815	79,323	79,323

Panel B: Recovery and Litigation				
	Recovery		Litigation	
	(1)	(2)	(3)	(4)
Domestic Bank Dummy	17.25 (6.70)	25.06 (4.95)	-0.23 (0.07)	-0.04 (0.08)
Non-Asian Foreign Bank Dummy	-28.53 (17.67)	-54.28 (11.65)	0.45 (0.10)	0.22 (0.13)
Log of Bank Size	4.17 (3.82)	1.30 (6.50)	0.01 (0.04)	0.02 (0.10)
Log of Bank Size * Big	1.30 (5.21)	8.23 (6.81)	0.04 (0.04)	0.03 (0.10)
Big	25.5 (72.65)	-110.31 (87.39)	0.49 (0.58)	0.41 (1.21)
Loan-Size Weighted	No	Yes	No	Yes
Other controls	Yes	Yes	Yes	Yes
Observations	5,762	5,762	5,762	5,762
R ²	0.20	0.25	0.22	0.13

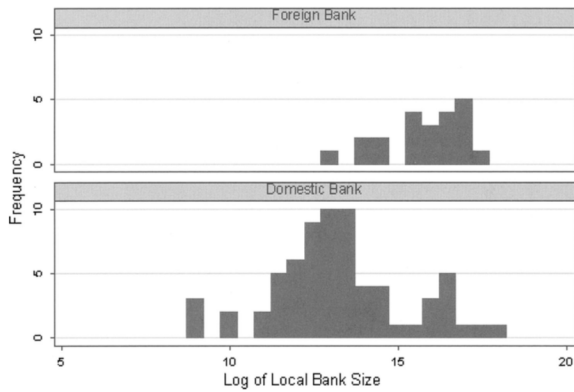


Figure 4a. Distribution of local bank size. The top panel draws the frequency histogram for the distribution of foreign banks by their local bank size. Bank size is defined as the total asset for a bank's operations in Pakistan. The lower panel does the same for domestic banks.

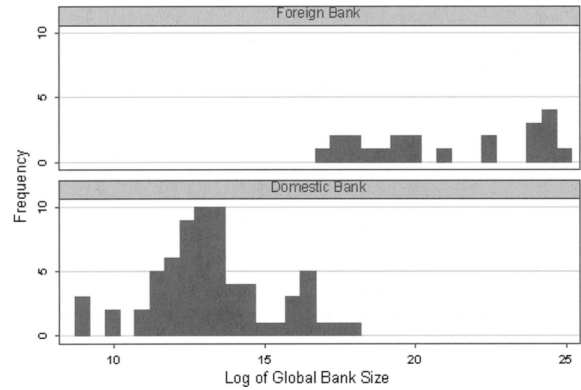


Figure 4b. Distribution of global bank size. The top panel draws the frequency histogram for the distribution of foreign banks by their global bank size. Global bank size is defined as the total assets for a bank's world-wide operations (data source: BankScope). The lower panel does the same for domestic banks.

Economic message. The main mechanism is not just that foreign banks come from different legal or regulatory systems. The evidence more directly supports cultural, communication, and agency costs.

5.11 Table IX: Are foreign-bank borrowers better-quality firms?

Read:

- The paper uses a separate export data set for textile firms to measure real borrower quality.
- Outcomes include whether the firm exports, log total exports, and export productivity.
- Firms borrowing only from foreign banks are not significantly better than firms borrowing only from domestic banks.
- Firms borrowing from both foreign and domestic banks tend to be larger or higher-quality, consistent with multiple-bank borrowing by stronger firms.

Economic message. The domestic-bank borrowers are not obviously low-quality firms that foreign banks correctly avoid. Domestic banks appear to serve viable firms that are informationally harder to finance.

5.12 Related-lending evidence

Read:

- The paper uses director information to identify loans where the borrower and domestic bank are related through directors or close family relationships.
- Only about **4.2%** of domestic-bank lending is classified as related, far below related-lending magnitudes documented in some other emerging-market settings.
- Related loans do not have significantly different default rates.

Economic message. The results are not driven by domestic banks lending to insiders or connected friends. Domestic private banking in this setting appears more disciplined than the risk-preference story suggests.

5.13 Table X: New branch lending and economic importance

Read:

- The paper studies loans from new branches opened during the sample period.
- New domestic branches make **7,104 loans**, compared with **1,611** by Asian foreign branches and **338** by non-Asian foreign branches.
- About **78.2%** of new domestic-branch loans go to first-time bank borrowers.

Table IX Are Foreign Bank Borrowers Better Quality Firms?						
The sample in this table is limited to firms in the textile sector. "Export?" is an indicator variable for whether the firm is an exporter, "Log Export" is the log of total exports for a firm from 2001 to 2003, and "Export Productivity" is total exports during 2001–2003 divided by total lending for the firm during the sample period. The data were collapsed at the firm (borrower) level, and so each observation represents a unique textile firm. The controls include 5 borrower size dummies, a dummy for whether the borrower is a foreign firm, 3 group size dummies, and 134 dummies for each of the city/town of borrower. Standard errors are clustered at the bank level.						
	Export?		Log Export		Log Export Productivity	
	(1)	(2)	(3)	(4)	(5)	(6)
Borrowing from both Foreign and Domestic	0.12 (0.04)	0.08 (0.03)	1.74 (0.54)	1.12 (0.43)	0.02 (0.17)	0.15 (0.12)
Borrowing from Foreign only	0.01 (0.04)	0.004 (0.02)	0.09 (0.46)	0.04 (0.30)	0.02 (0.17)	−0.004 (0.12)
Constant	0.23 (0.03)	—	2.75 (0.42)	—	0.98 (0.13)	—
Controls	Yes		Yes		Yes	
R ²	0.01	0.09	0.01	0.1	0.00	0.07
Observations	4,573	4,573	4,573	4,573	4,573	4,573

**Table X
Lending Composition of New Bank Branches Opened after 1996**
Data in Panels A and B are restricted to loans given out by "new" bank branches during the sample period. These are branches that were opened during the period June 1996 to April 2002. Panel B reports differences in lending composition between domestic, Asian foreign, and non-Asian foreign banks. Size is coded 1 through 5, representing the five borrower-size classifications shown in Table II; Location is coded 1 through 3, representing the city size of the borrower as in Table II; Foreign Firm is a dummy for whether the borrower is a multinational; Group Size is coded 1 through 3, representing the group size of the borrower; and Loan Type is a dummy for whether the loan is a short-term loan. Standard errors in parentheses are clustered at the bank level (90 banks in all).

Panel A: Previous Banking Experience of Borrowers						
	Domestic Banks		Asian Foreign Banks		Non-Asian Foreign Banks	
	No. of Loans	Percentage of Loans	No. of Loans	Percentage of Loans	No. of Loans	Percentage of Loans
None (First Time Bank Borrower)	5,555	78.2	1,022	63.4**	130	38.46**
Only Government Banks	471	6.6	115	7.1	4	1.2
Includes Domestic Banks	627	8.8	273	17.0	11	3.3
Includes Foreign Banks	222	3.1	72	4.5	144	42.6
Includes Domestic and Foreign Banks	229	3.2	129	8.0	49	14.5
<i>Total</i>	<i>7,104</i>		<i>1,611</i>		<i>338</i>	

Panel B: Value-Weighted Means for "Hard Information" Proxies						
	Size		Foreign Firm	Group Size	No. of Creditors	Loan Type
		Location				
Domestic Banks	3.57 (0.17)	2.47 (0.19)	0.00 (0.00)	1.87 (0.07)	1.99 (0.30)	0.95 (0.04)
Asian Foreign Banks	4.23 (0.25)	2.34 (0.12)	0.00 (0.00)	2.34 (0.24)	3.13 (0.23)	0.95 (0.05)
Non-Asian Foreign Banks	4.28 (0.19)	2.98 (0.01)	0.05 (0.04)	2.07 (0.27)	3.14 (0.25)	0.99 (0.01)

**63.4 and 38.5 are both statistically different from 78.2 (the percentage of first time borrowers from domestic banks) at 10% and 5% significance levels respectively, with standard errors clustered at the bank level.

- The corresponding shares are **63.4%** for Asian foreign banks and **38.46%** for non-Asian foreign banks.
- New domestic branch lending is softer than new foreign branch lending: borrowers are smaller, less likely to be foreign, and have fewer creditors.

Economic message. Distance constraints are economically large. New domestic banks do not merely poach soft borrowers from foreign banks; they extend credit to borrowers that foreign banks had not served.

6 Short summary

- The paper studies whether foreign banks can substitute for domestic banks in a poor economy.
- It uses detailed loan-level data from Pakistan from 1996 to 2002.
- Foreign banks and private domestic banks have similar aggregate private-bank market shares, but lend to different borrowers.
- Foreign banks lend heavily to hard-information borrowers: large firms, large cities, foreign firms, conglomerates, and firms with multiple creditors.
- Domestic banks lend more to soft-information borrowers that require relationship lending.
- Conditional on default, domestic banks recover more and renegotiate more successfully.
- Borrower fixed effects show that the recovery gap survives even for the same borrower.

- Non-Asian foreign banks are more constrained than Asian foreign banks, supporting a geographic and cultural distance mechanism.
- The paper rejects alternative explanations based on political enforcement, bank size, institutional distance, poor domestic-bank loan quality, borrower productivity, and related lending.
- New domestic branches mostly lend to first-time borrowers, showing that domestic banks expand access to credit rather than simply replacing foreign lenders.

7 Conclusion

7.1 Core conclusion

Foreign banks are not a complete solution to financial underdevelopment. They can provide capital and discipline, but cultural, geographic, and organizational distance limits their ability to lend to firms that require soft information and relationship-based contracting.

7.2 Economic intuition

Soft-information lending requires local knowledge, trust, discretionary judgment, renegotiation ability, and monitoring. These tasks are hard to perform when the relevant information must be communicated through a distant foreign banking organization. Domestic banks are closer to the local social, business, and informational environment, so they can finance borrowers that foreign banks avoid.

7.3 Takeaway

The paper's policy implication is balanced. Foreign banks can improve banking quality and stability, but poor economies cannot rely only on foreign banks to deepen credit access. A strong domestic financial sector is essential for financing small, local, opaque, and relationship-dependent firms.